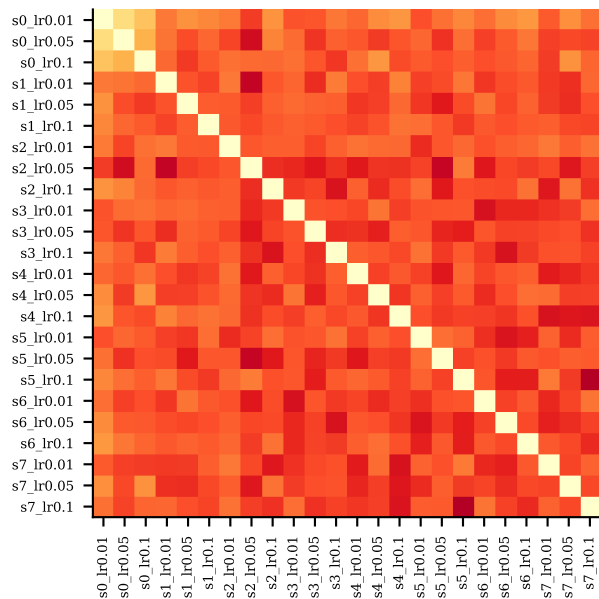
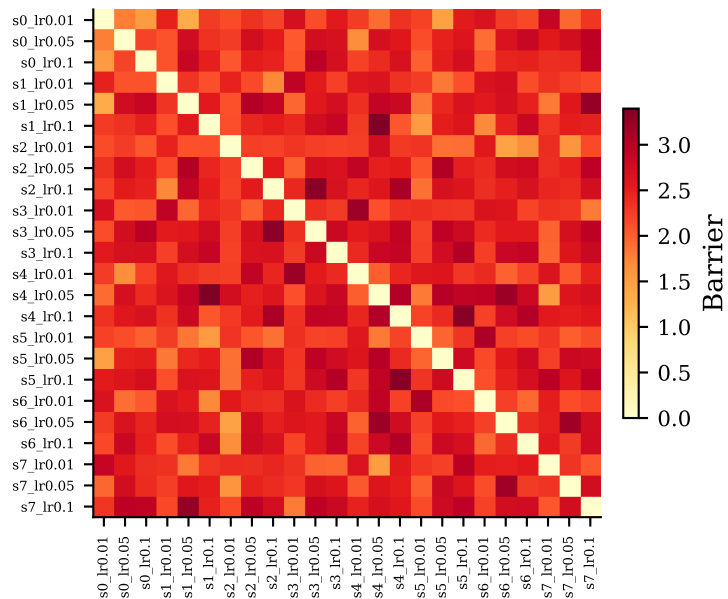


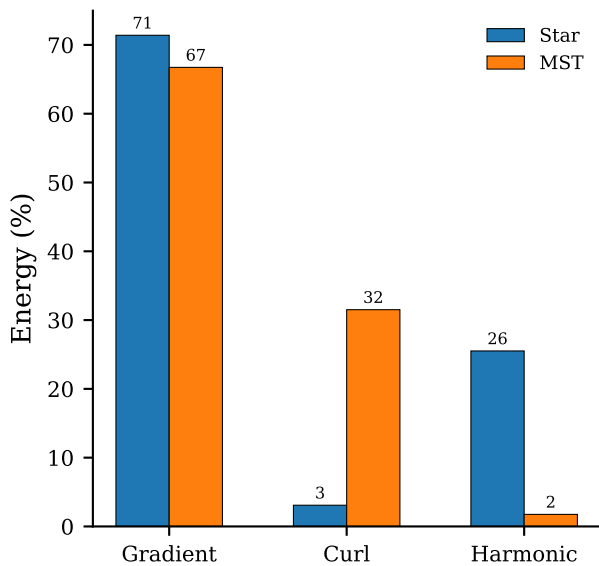
(a) Star graph barriers



(b) MST graph barriers



(c) Hodge decomposition

(d) Loop complexity β_1 